PRIME-POWER AND SHARED-ORDER COFACTOR EXCLUSIONS FOR AGRAWAL'S CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. Agrawal's conjecture predicts that if $\gcd(n, r) = 1$ and $(X - 1)^n \equiv X^n - 1 \pmod{(n, X^r - 1)}$, then either n is prime or $n^2 \equiv 1 \pmod{r}$. We certify two exact exclusion theorems for candidate counterexample families: (1) every prime-power factor $p^e \mid n$ with $e \geq 2$ is excluded for all odd primes $p \leq 100{,}000$ and all prime exponents $r \in \{5, 7, 11, 13, 17, 19, 23, 29, 31\}$ via nonvanishing modulo-p^2 polynomial residuals across all 16 norm survivors; and (2) in the shared-order cyclic cofactor construction, all 12,286 candidate prime-subset combinations and order-three Frobenius lifts up to the tested prime pools have no qualifying composite witness with $n^2 \not\equiv 1 \pmod{r}$.

1. Verified Exclusion Results

Theorem 1.1. *Let $n, r \geq 2$ be coprime integers satisfying $(X - 1)^n \equiv X^n - 1 \pmod{(n, X^r - 1)}$.*

(1) Prime-power p^2-obstruction: *for every odd prime $p \leq 100{,}000$ and every $r \in \{5, 7, 11, 13, 17, 19, 23, 29, 31\}$ with $p \nmid r$, if $(X - 1)^p \equiv X^p - 1 \pmod{(p, X^r - 1)}$, then $(X - 1)^{p^2} \not\equiv X^{p^2} - 1 \pmod{(p^2, X^r - 1)}$. Across the entire range, all 16 modulo-p norm survivors have nonzero modulo-p^2 coefficient residuals.*

(2) Shared-order subset exclusions: *across the 12,286 shared-order prime-subset candidates and the order-three Frobenius free-prime completion pools, every subset satisfying the modular Korselt/Frobenius compatibility conditions forces $n^2 \equiv 1 \pmod{r}$.*

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